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Fayle et al.

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(54) **CIRCUMFERENCE CLOSING SYSTEM**

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Sep. 23, 2021, now Pat. No. 11,930,891, which is a
continuation of application No. 15/654,533, filed on
Jul. 19, 2017, now Pat. No. 11,147,328.

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(52) **U.S. Cl.**

CPC **B65D 33/28** (2013.01); **A41F 9/025**
(2013.01); **A41F 1/00** (2013.01); **A41F 9/02**
(2013.01)

(58) **Field of Classification Search**

CPC ... **B65D 33/28**; **A41F 1/00**; **A41F 9/02**; **A41F**
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USPC **383/71**, **72**, **75**; **24/30.5 R**
See application file for complete search history.

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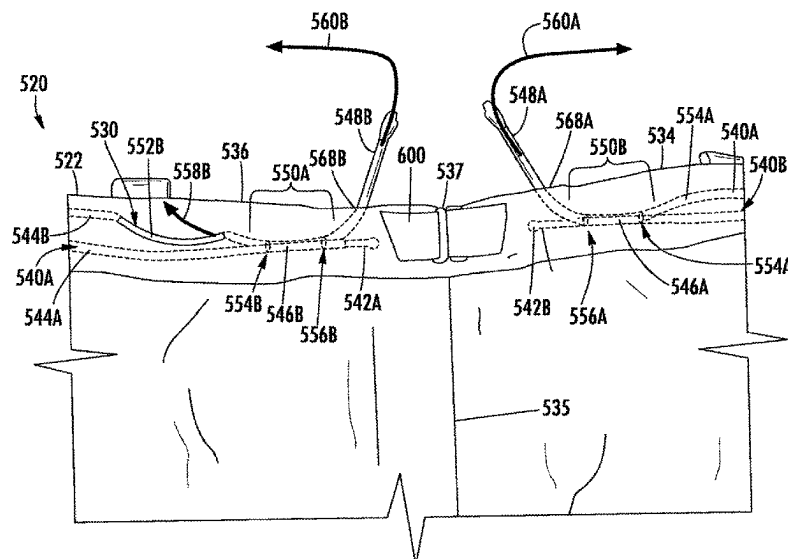
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(57)

ABSTRACT

Various example garment, footwear and bag articles may
include a closable circumference in the form of a mouth,
waist or foot opening. A circumference closing system may
utilize at least one constrict-able tubular length to releasably
retain a selected size of the closable circumference or
opening.

12 Claims, 10 Drawing Sheets

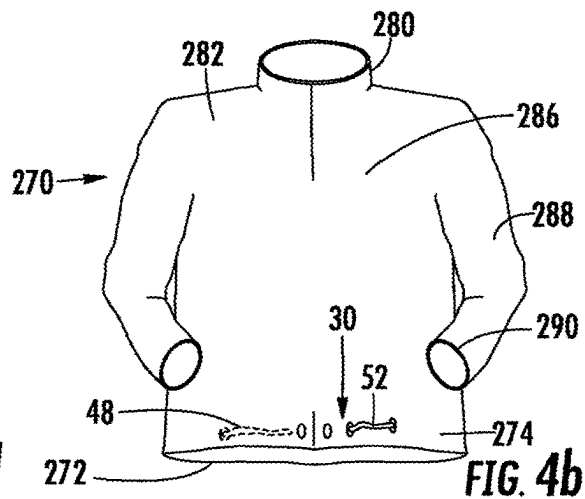
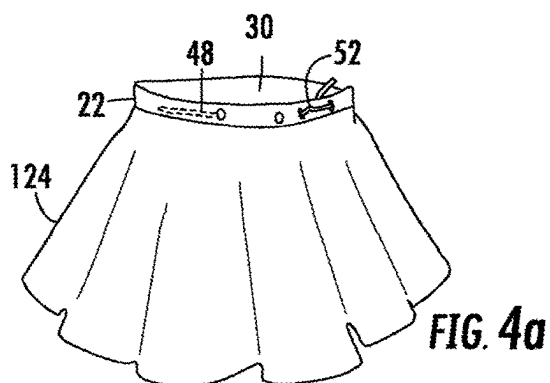
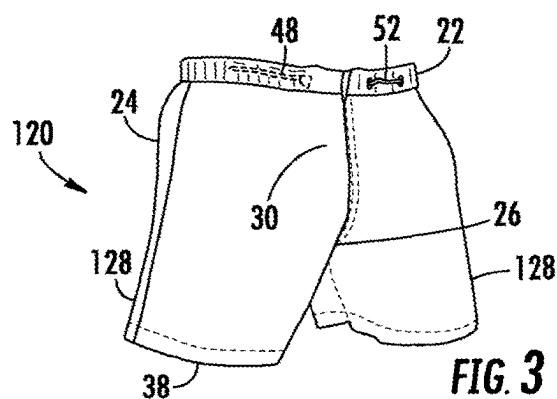
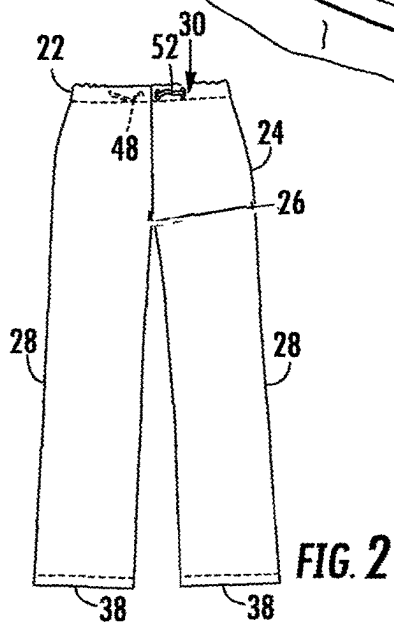
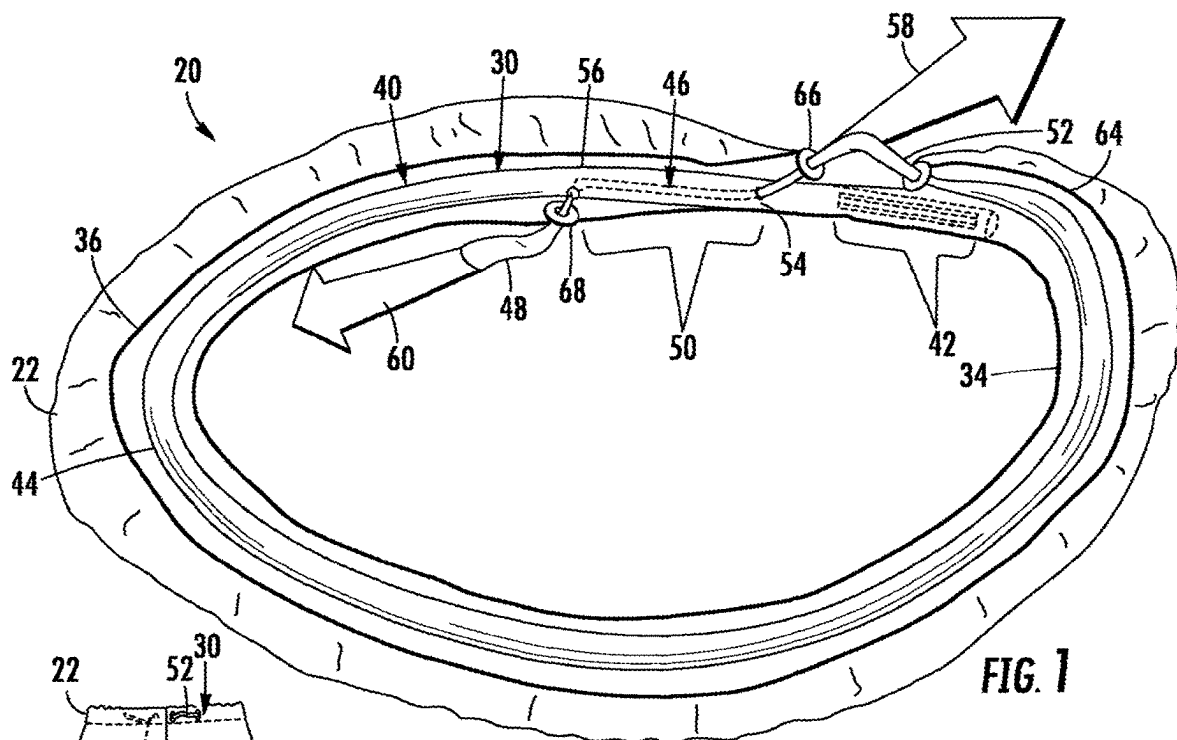


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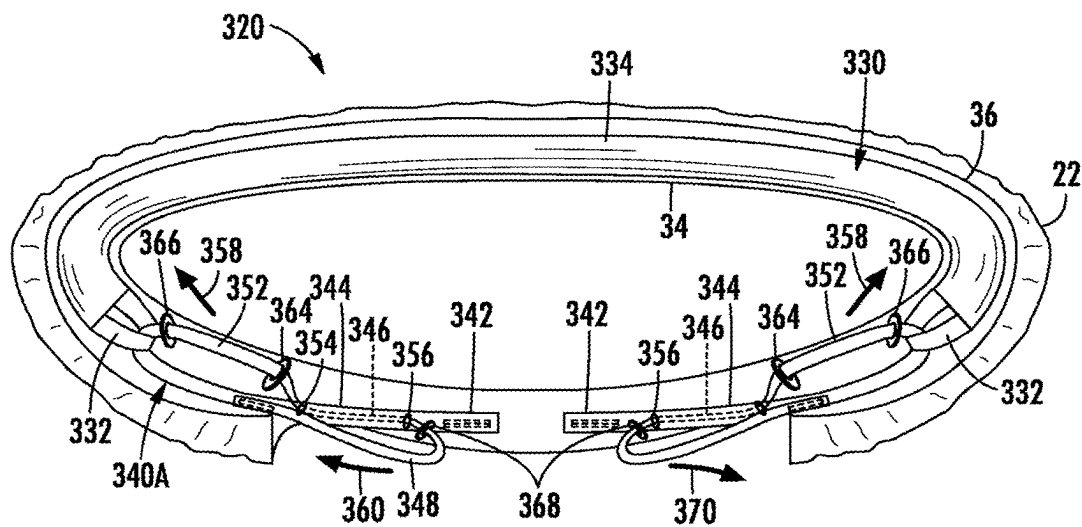


FIG. 5a

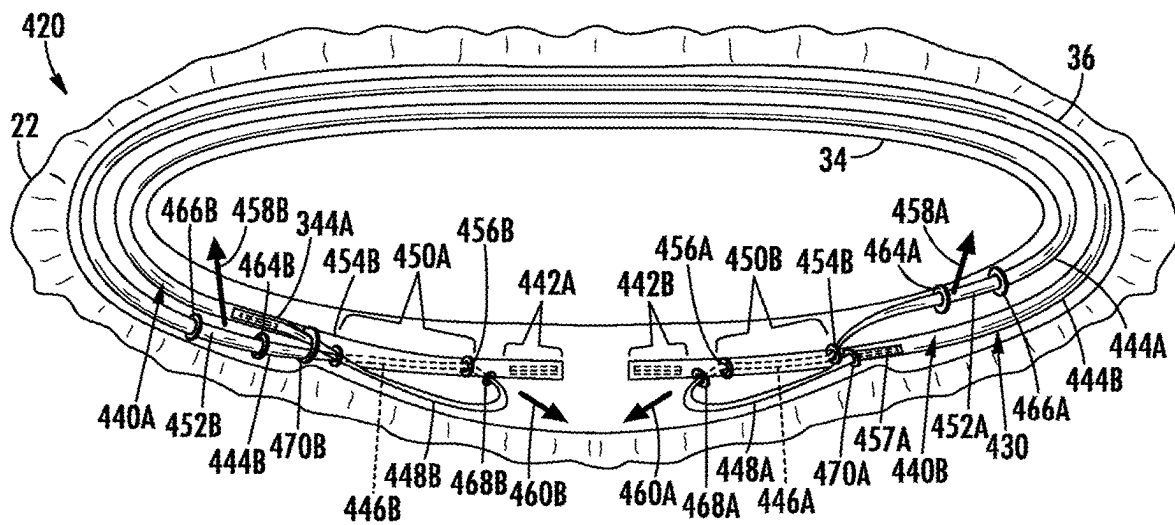


FIG. 6a

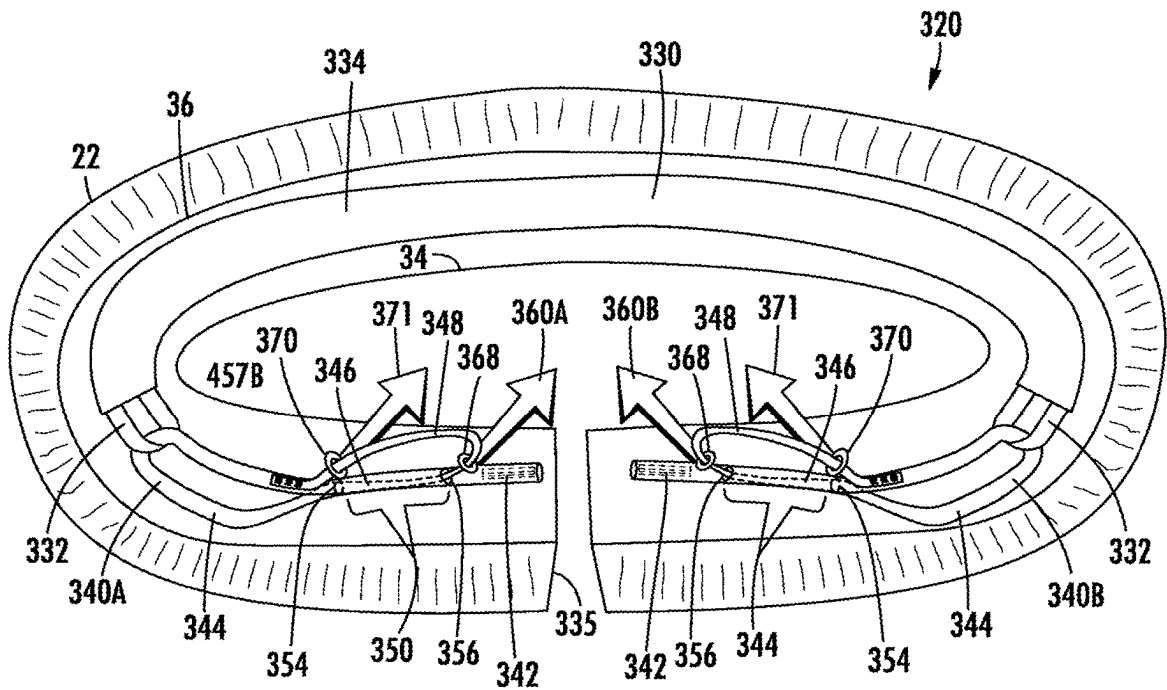


FIG. 5b

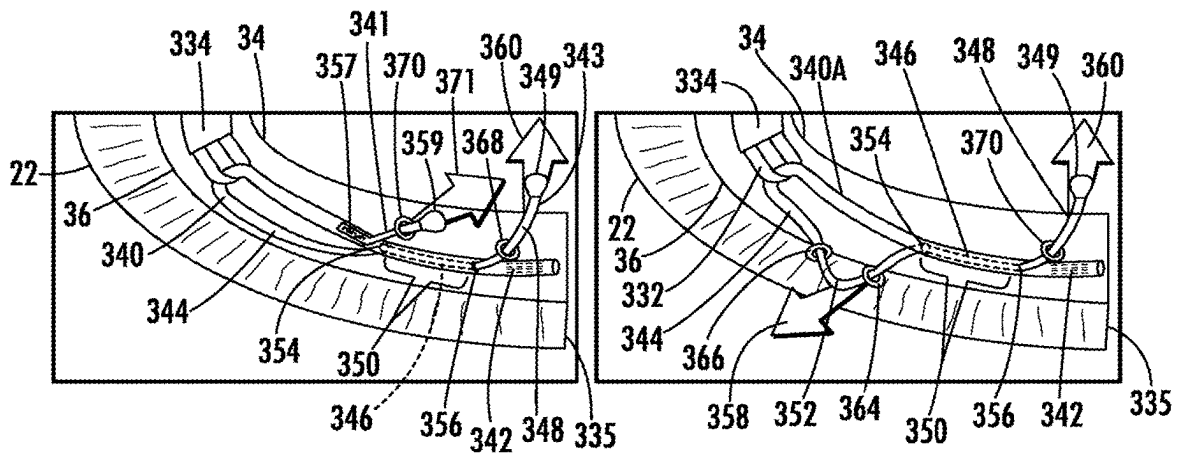


FIG. 5c

FIG. 5d

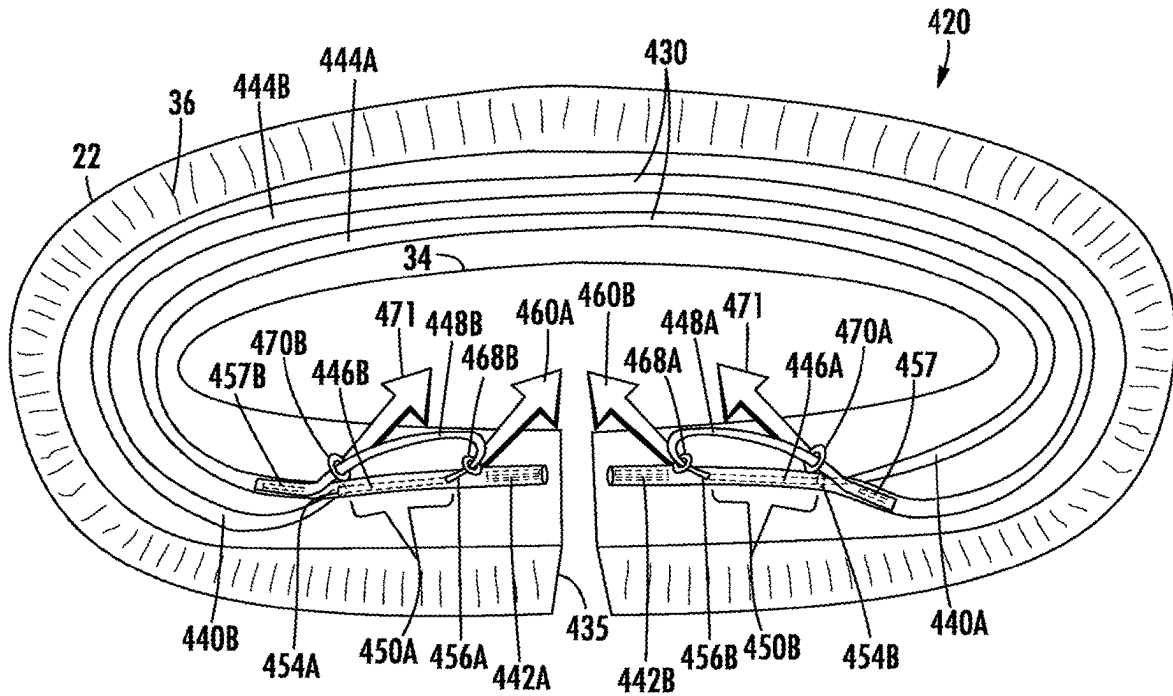


FIG. 6b

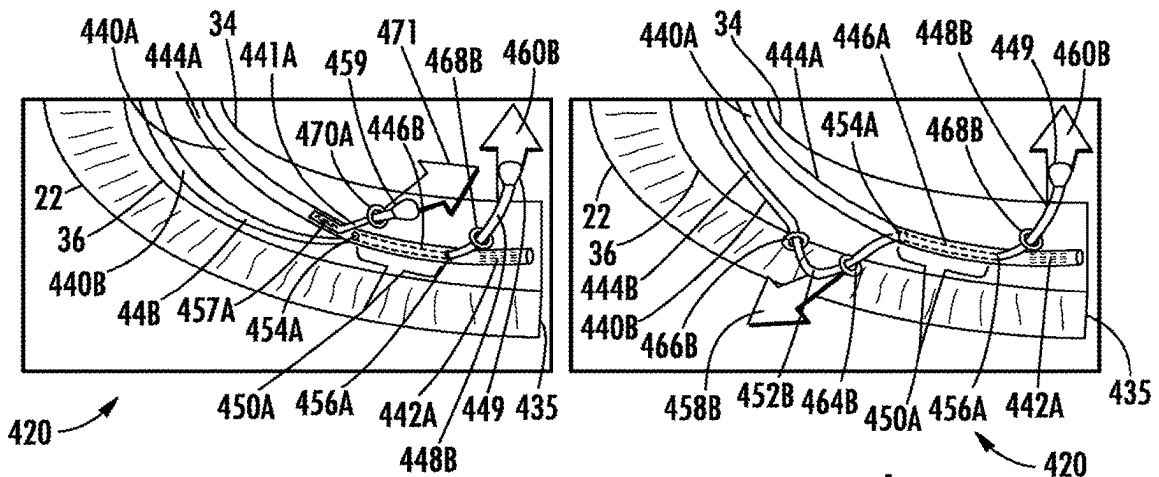
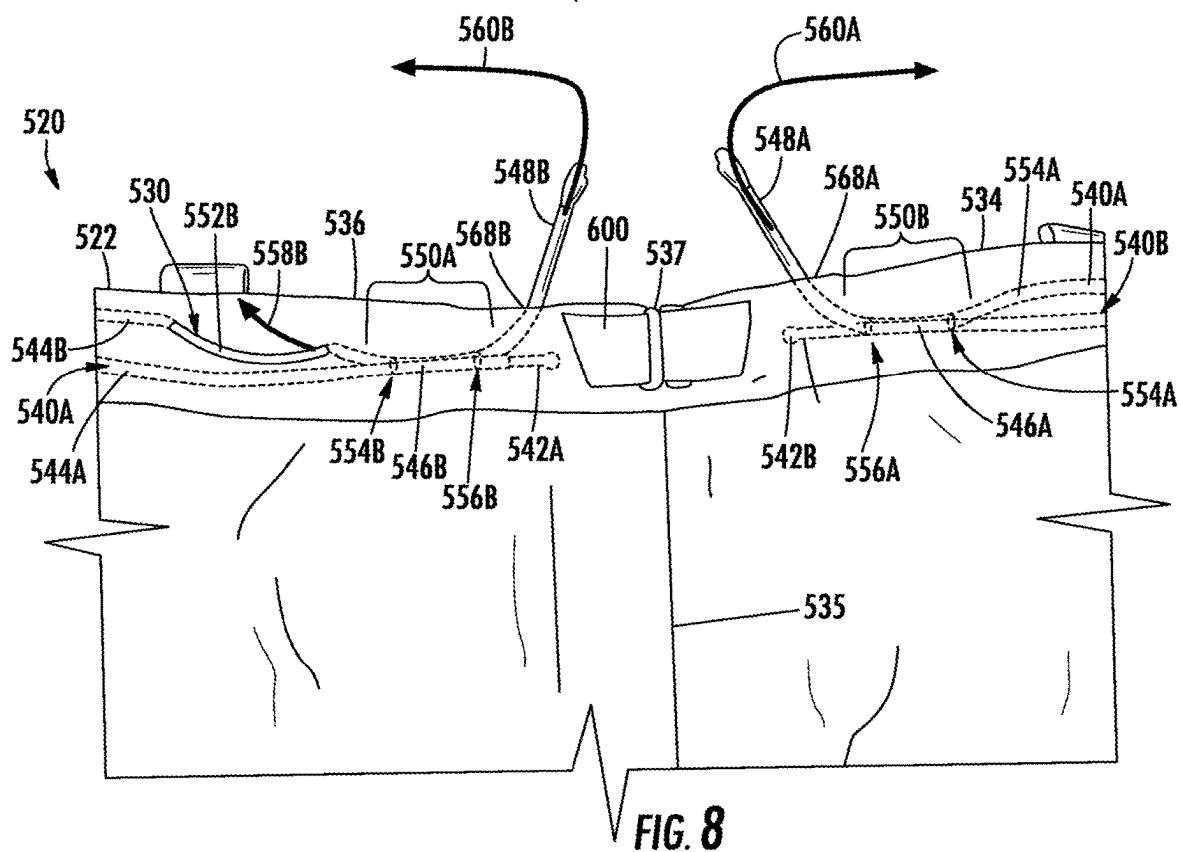
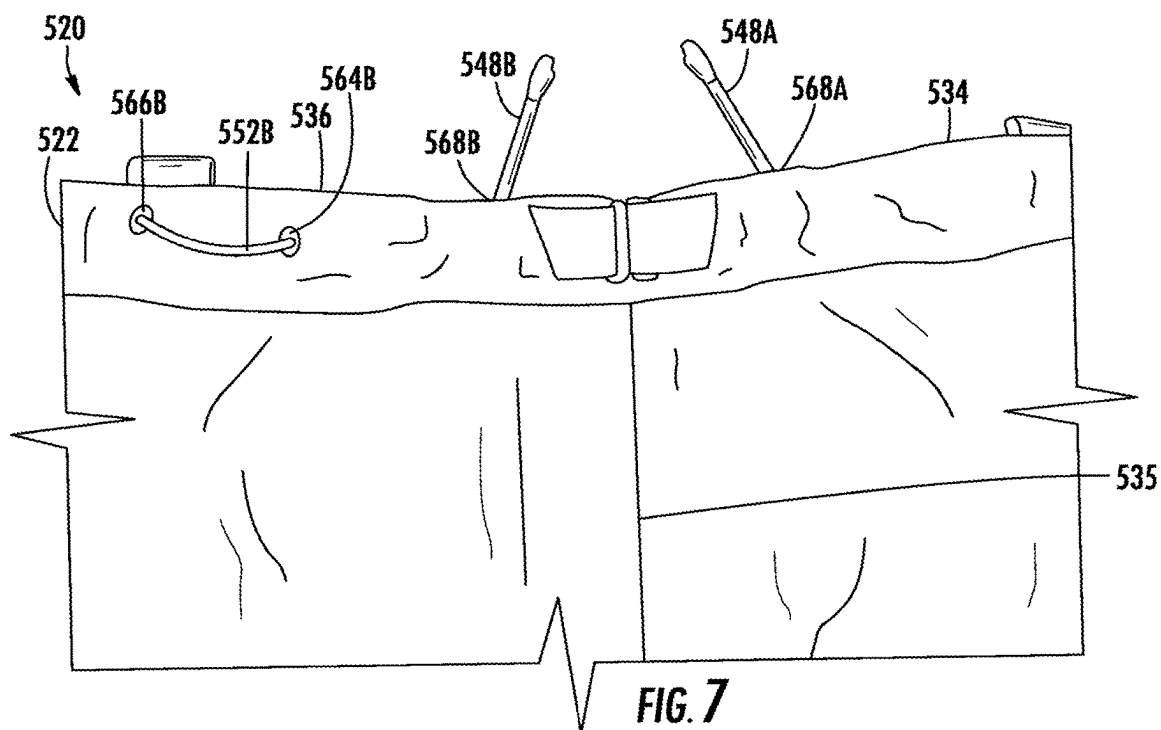
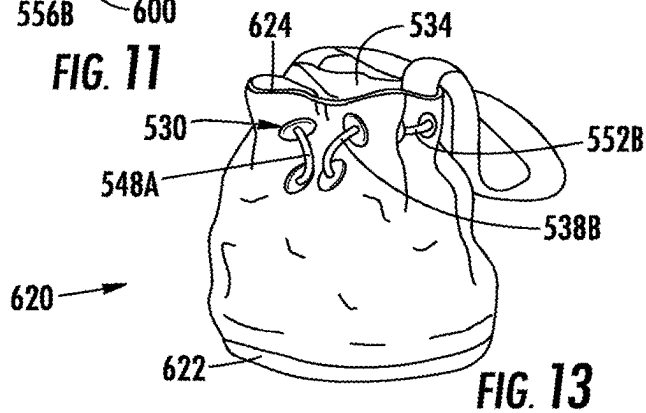
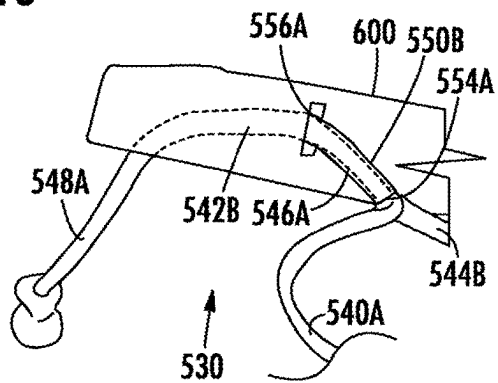
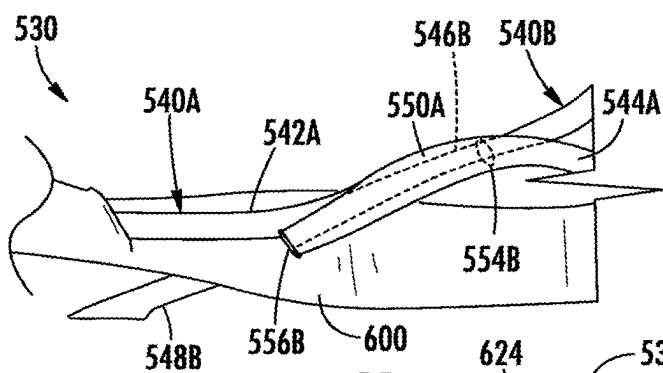
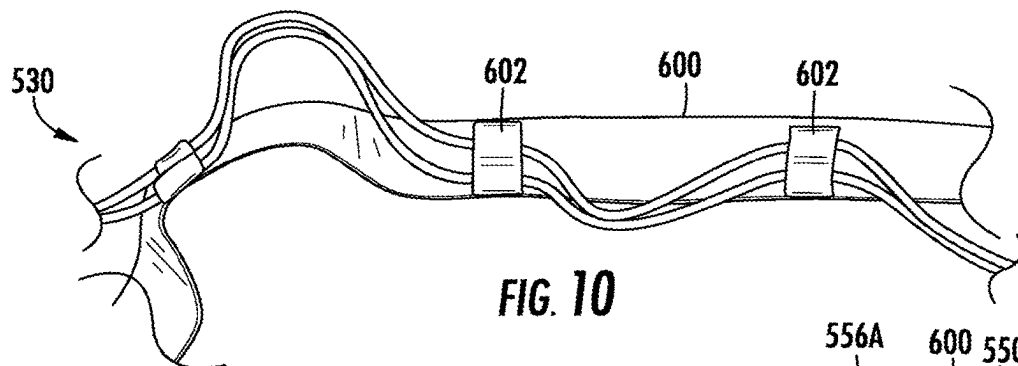
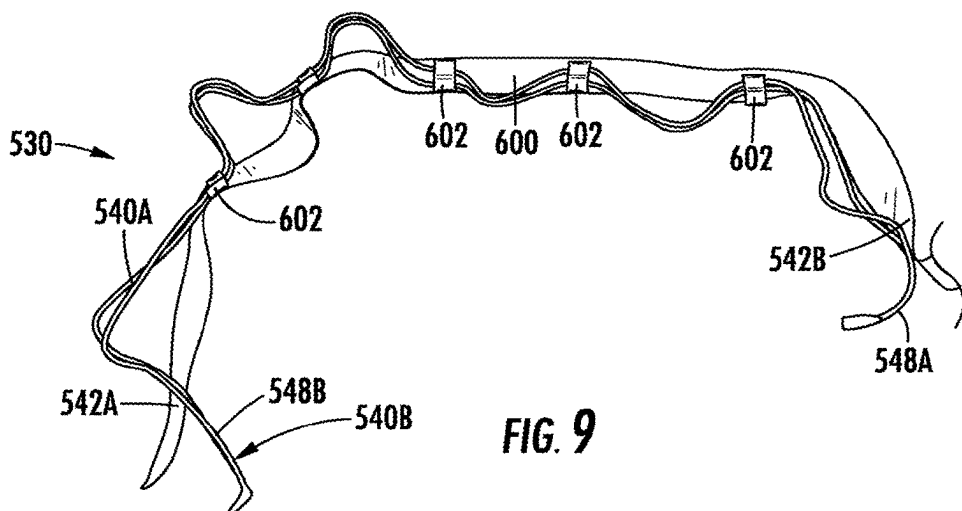


FIG. 6c

FIG. 6d





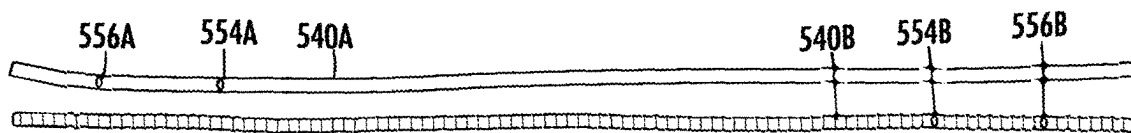


FIG. 14

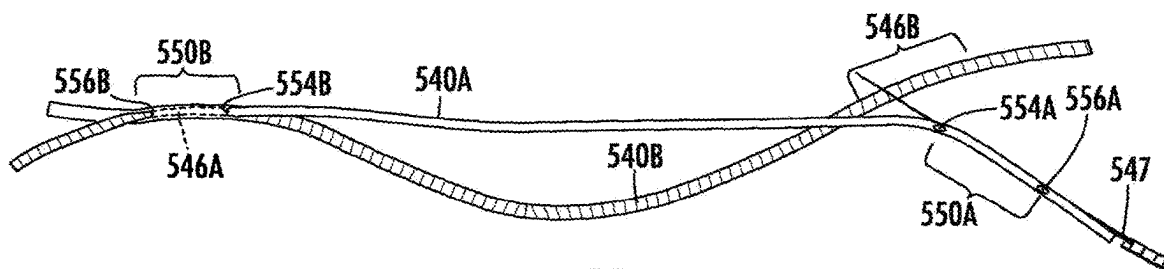


FIG. 15

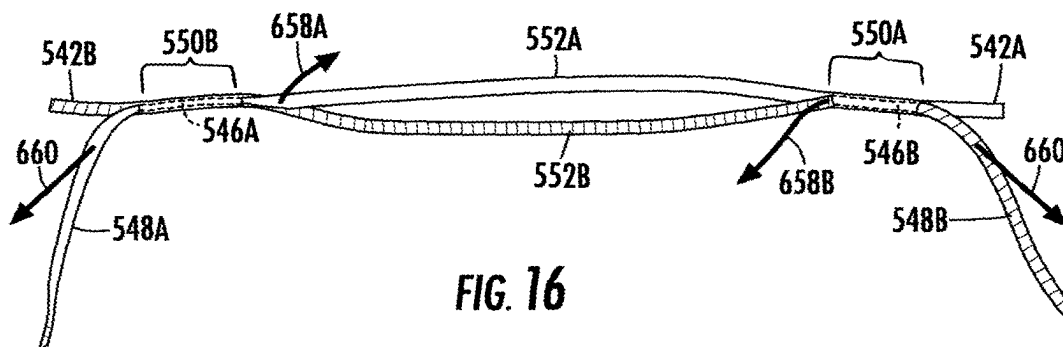


FIG. 16

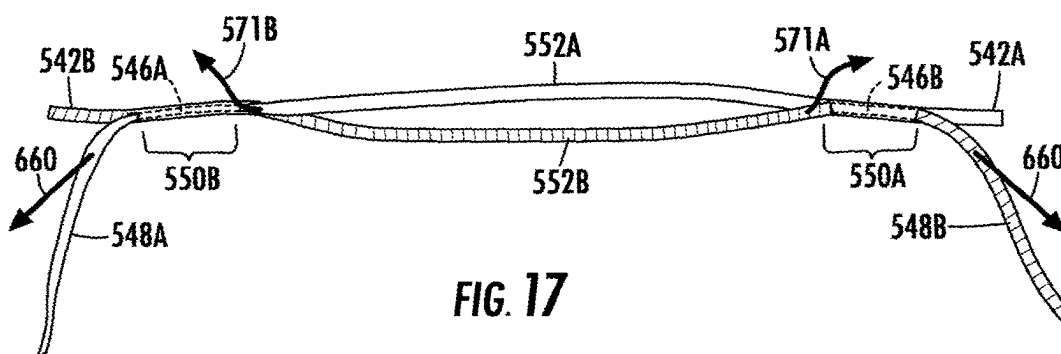


FIG. 17

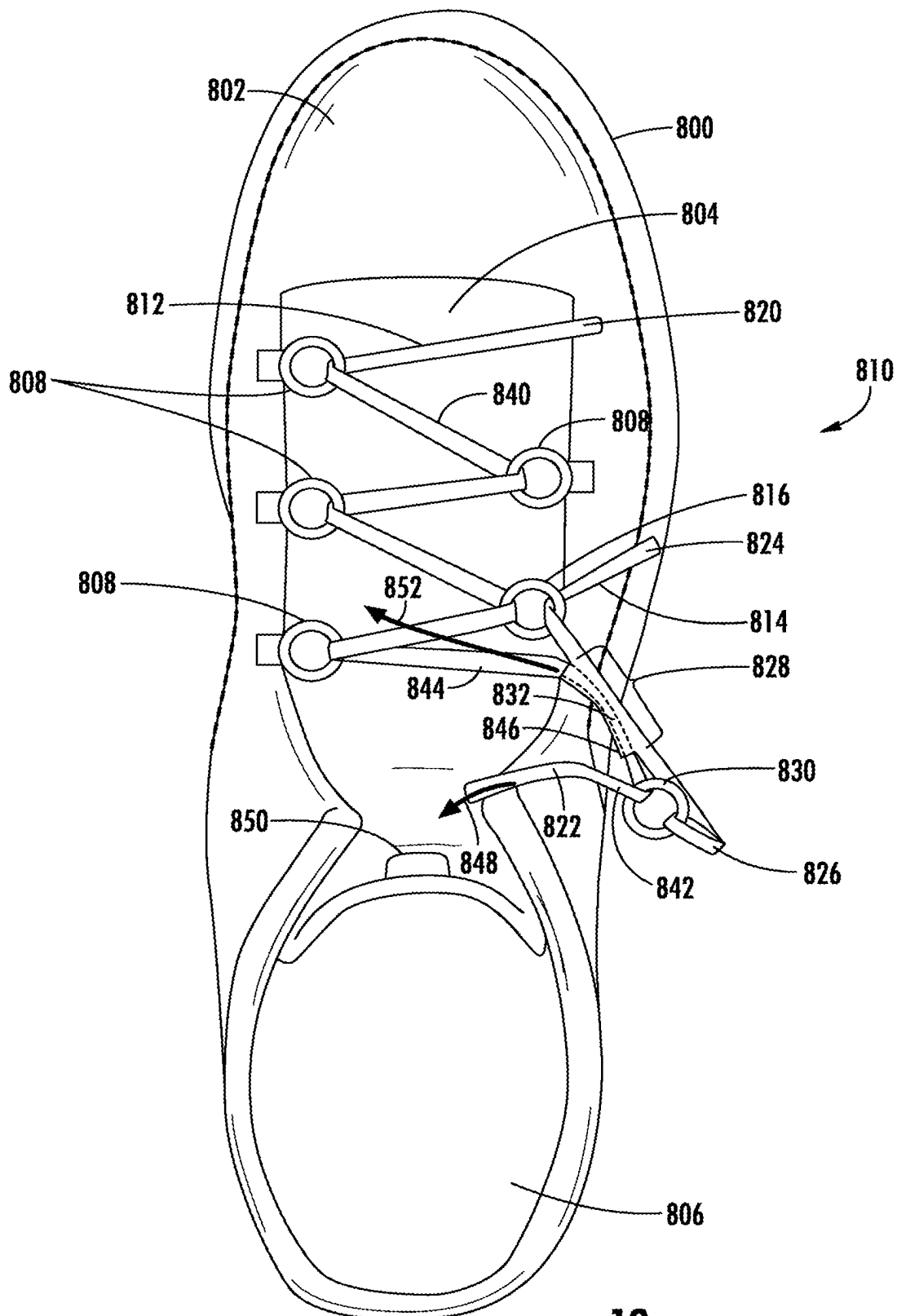


FIG. 18

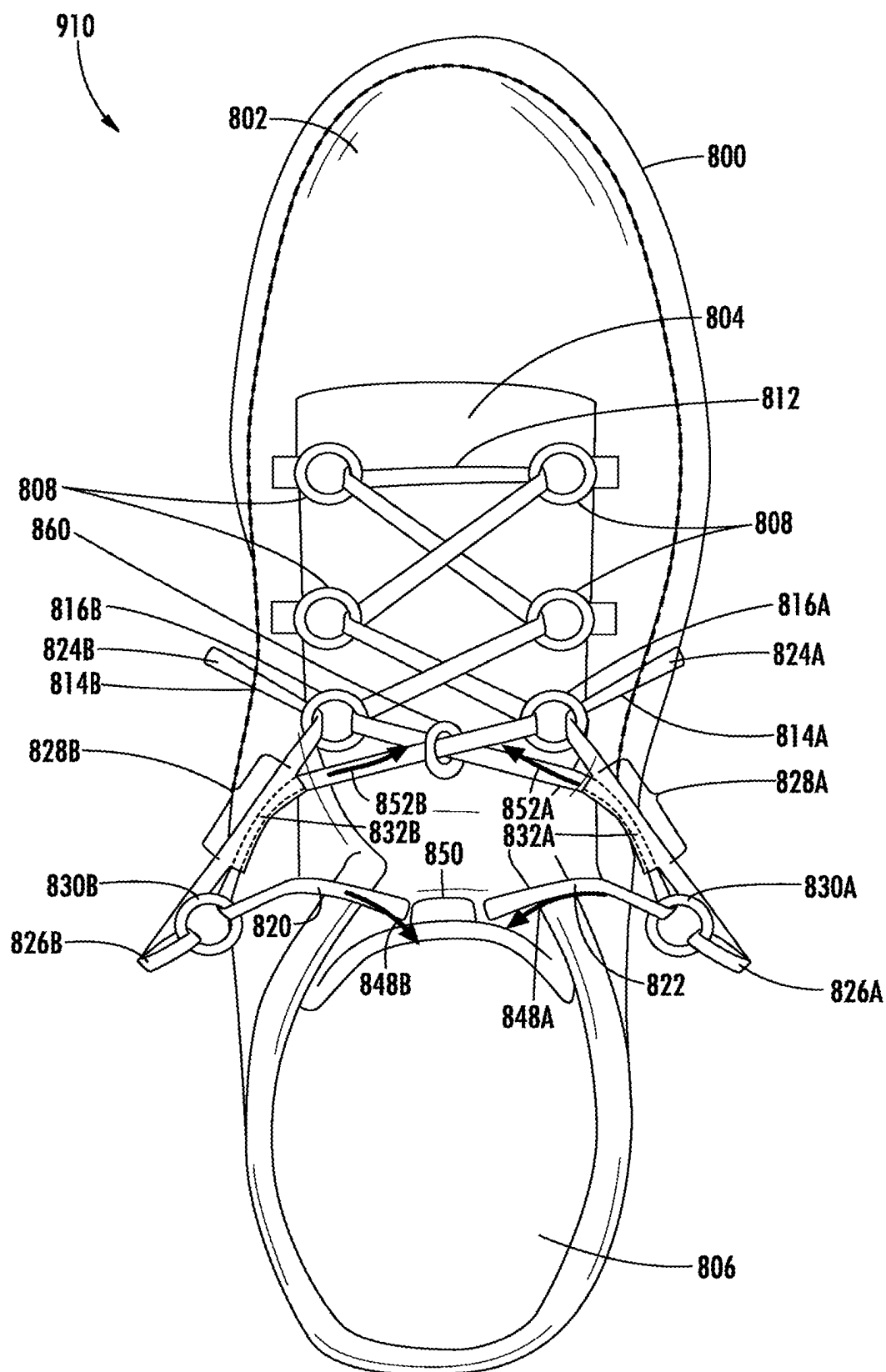


FIG. 19

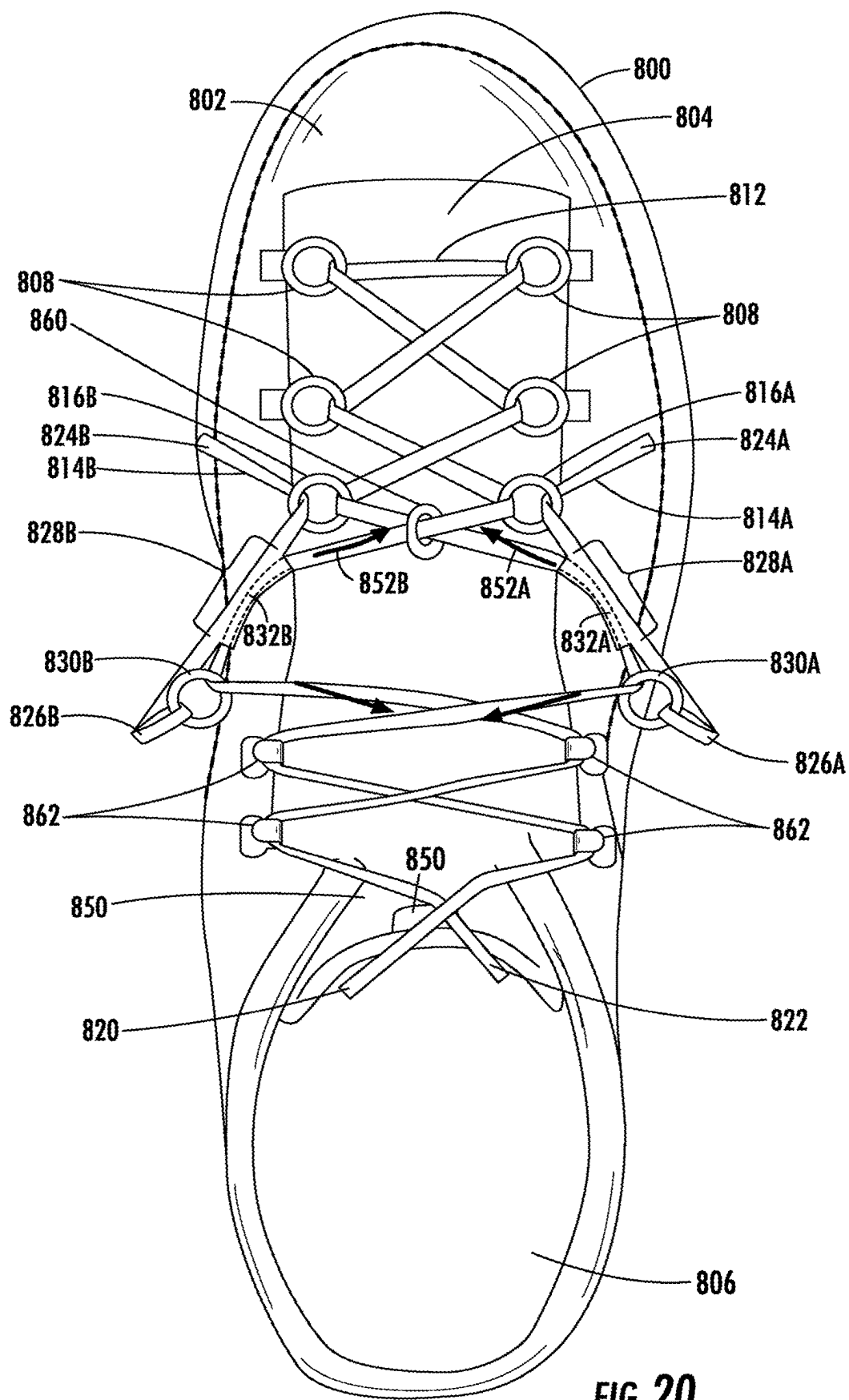


FIG. 20

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CIRCUMFERENCE CLOSING SYSTEM**CROSS REFERENCE TO RELATED APPLICATION**

The present application is a continuation application claiming priority under 35 USC § 120 from U.S. patent application Ser. No. 17/482,859 filed on Sep. 23, 2021, which is a continuation application of U.S. patent application Ser. No. 15/654,533 filed on Jul. 19, 2017, now U.S. Pat. No. 11,147,328, the full disclosures of which are hereby incorporated by reference.

BACKGROUND

Many pieces of apparel or garments are worn about a person's waist and have a closable waist, a waist that may be constricted to adapt to individual waist sizes as well as personal preferences. Other garments extend over a person's torso and can include a constrictable bottom opening or mid-section, such as draw strings. Such garments come in a variety of different forms such as running shorts, swim shorts, sweatpants, slacks, skirts, bibs, jackets, coats, vests, hoodies, pullovers, sweatshirts, shirts, and the like. Footwear articles have a variety of different lacing or closing systems. Various mechanisms are currently used to control the constriction of the closable waist or the foot opening of footwear, such as a belt, hook and loop fasteners, tri-glides, webbings, snaps, clips, laces and the like. Such mechanisms may fail, slide or loosen when unloaded or slide or loosen under load. Moreover, such mechanisms may add undesirable hardware, may be weak, may add undesirable amounts of weight, may limit movement, may be difficult to use, or may increase the bulk of the closable waist of the garment.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a fragmentary top view of a portion of an example garment that incorporates an example circumference closing system.

FIG. 2 is a front view of the example garment of FIG. 1.

FIG. 3 is a front perspective view of another example garment incorporating the example circumference closing system.

FIGS. 4a and 4b are front perspective views of other example garments incorporating the example circumference closing system.

FIGS. 5a through 5d are fragmentary top views of portions of example garments incorporating other example implementations of circumference closing systems.

FIGS. 6a through 6d are fragmentary top views of portions of example garments incorporating other example implementations of circumference closing systems.

FIG. 7 is a fragmentary front view of a portion of another example garment incorporating another example circumference closing system.

FIG. 8 is a fragmentary view of the portion of the example garment with portions transparently illustrated are shown in broken lines to further illustrate the example circumference closing system.

FIG. 9 is a top perspective view of the example circumference closing system separated from a remainder of the garment.

FIG. 10 is an enlarged view of a portion of the circumference closing system of FIG. 9.

FIG. 11 is an enlarged view of another portion of the circumference closing system of FIG. 9.

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FIG. 12 is an enlarged view of another portion of the circumference closing system of FIG. 9.

FIG. 13 is a perspective view of an example bag incorporating the example circumference closing system of FIG. 9.

FIGS. 14-17 are front views illustrating an example method for forming a portion of the example circumference closing system of FIG. 9.

FIG. 18 is a top view of an article of footwear incorporating an example implementation of a footwear closing system.

FIGS. 19 and 20 are top views of articles of footwear incorporating other example implementations of footwear closing systems.

DETAILED DESCRIPTION OF EXAMPLES

Disclosed herein are example garments that provide a closable circumference or waist and that utilize a circumference closing system that utilizes at least one constrictable tubular length to releasably retain a selected size of the closable circumference. The disclosed circumference closing system may be used to supplement or replace existing mechanisms or hardware in a variety of garments and other applications. The disclosed circumference closing system may eliminate undesirable hardware, may reduce the weight of the garment, may increase movement or flexibility of the garment or may reduce the bulk of the closable waist of the garment.

FIGS. 1 and 2 illustrate an example garment 20, shown as a generic pair of pants. FIG. 1 is a top view of a portion of pants 20 while FIG. 2 is a front view of pants 20. Pants 20 may have various forms and configurations and go by various name other names such as slacks, sweatpants, trousers, tights, jeans, gauchos, pedal pushers, palazzos, pajamas, jodhpurs, knickers, bibs and the like.

As shown by FIG. 2, pants 20 generally comprises waist portion 22, hips portion 24, crotch 26, leg portions 28 and circumference closing system 30. Waist portion 22 comprise a body of material, such as woven material providing a constricted or constrictable opening 34 that receives a person's waist and that supports the remaining portions of pants 20 when being worn by a person. As shown by FIG. 1, in the example illustrated, waist portion 22 has an internal sleeve 36 that receives portions of circumference closing system 30.

In one implementation, waist portion 22 provides a continuous uninterrupted body of material that continuously extends about a person's waist. One example of such an implementation is pants having an uninterrupted body of material that continuously extends about a person's waist is a sweatpants. In another implementation, waist portion 22 may be split or have a front zipper, wherein opposite sides of the split are temporarily clasped, buttoned or otherwise secured to one another. In other examples, pants can include one or more side zipper or reclosable slits or splits. In all such cases, waist portion 22 is flexible and/or compressible so as to be constrictable, facilitating adjustment of the size of waist portion 22 through the use of circumference closing system 30.

Hips portion 24 comprise that portion of pants 20 extending from waist portion 22 to crotch 26. Hips portion 24 extends about a person's hips when pants 20 is worn. In some implementations, hips portion 24 may have at least one pocket secured thereto on the front or sides.

Leg portions 28 extend from hips portion 24, beginning at crotch 26 and terminating at leg openings 38. The distance

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between crotch 26 and leg openings 38 is sometimes referred to as an inseam. Although illustrated as substantially straight, leg portions 26 may have a variety of shapes and configurations such as loose fit, tapered, bell bottom and the like.

Circumference closing system 30 engages waist portion 22 and is closable or constrict-able so as to adjust the circumference or the interior size of waist portion 22 to accommodate different size waists and/or other undergarments of users who may be wearing garment 20. Circumference closing system 30 comprises a single line 40 at least partially received within sleeve 36. Line 40 comprises portions 42, 44, 46, 48 and 50.

Portion 42 comprises a first end portion of line 40 secured to waist portion 22. In one implementation, portion 42 is stitched to waist portion 22. In another implementation, portion 42 is welded, bonded or otherwise secured to waist portion 22.

Portion 44 of line 40 extends from portion 42 and continues along waist portion 22, within sleeve 36, about substantially an entirety of the closable circumference of waist portion 22. In the example illustrated, portion 44 comprises a constrict-able tubular length 50 and an exposed portion 52. Constrict-able tubular length 50 comprises a length of line 40 which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length 50. In one implementation, the entire length of line 40 is a constrict-able tube. For example, in one implementation, the entire length of line 40 comprises a braided tube of material. In another implementation, the entire length of lines 40 comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length 50 may comprise a portion of line 40, wherein other portions of line 40 are not tubular or are not constrict-able.

Portion 46 comprises that portion of line 40 that is extending through the hollow interior of constrict-able tubular length 50, entering at entry 54 and exiting at exit 56. Portion 46 is sized to enable portion 46 to pass through the constrict-able tubular length 50. Portion 48 comprises that portion of line 40 that extends from exit 56 and from length 50. Portion 48 provides a segment of line 40 by which a person may grasp and pull line 40 to adjust a length of line 40 that extends from portion 42, about the waist of the person wearing garment 20, and to the constriction of length 50. In one implementation, portion 48 is a loose, unattached second end of line 40. In another implementation, an end of portion 48 may be secured to waist portion 22 so as to not hang or dangle.

Exposed portion 52 comprises a segment of portion 44 of line 40 that is exposed outside of waist portion 22 and sleeve 36 such that a person may grasp portion 52 to pull back portion 48 through length 50 (as indicated by arrow 58), shortening the exposed length of portion 48 and increasing the length of portion 44 extending between the constriction of length 50, about the waist of the person wearing garment 20 and back to portion 42. In other words, exposed portion 52 facilitates lengthening of portion 44 and widening of the closable circumference of waist portion 22.

In the example illustrated, sleeve portion 36 and waist portion 22 comprise openings 64, 66 and 68. In one implementation, each of opening 64, 66 and 68 may be defined or protected by a grommet or other lining. Opening 64 comprise an opening from which portion 44 exits sleeve 36 and waist portion 22. Opening 66 comprises an opening into

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which portion 44 reenters sleeve 36 and waist portion 44, further extending towards constrict-able tubular length 50. Openings 64 and 66 facilitate the provision of exposed portion 52, allowing line 44 to be loosened when the size of waist portion 22 is to be increased. Opening 68 comprise an opening in sleeve 36 and waist portion 22 through which portion 48 exits, facilitating grasping a portion 48 when waist portion 22 is to be constricted, closed or reduced in size.

When a person wishes to reduce or constrict the size of closable circumference of waist portion 42, the person may pull portion 48 through length 50 (as indicated by arrow 60), increasing the length of portion 48 and reducing the length of portion 44 extending between the constriction of length 50, about the waist of the person wearing garment 20 in back to portion 42. After the length of portion 44 in the size of the closable circumference of waist portion 22 have been established, circumference closing system 30 retains the chosen size of the closable circumference of waist portion 42. Outward forces exerted upon waist portion 42 and upon line 44 place an axial load on length 50 which constricts about and against portion 46. The constriction of portion 50 about portion 46 frictionally locks portion 50 about the entire circumference or about all sides of portion 46, inhibiting axial sliding movement of portion 46 within the hollow interior of constrict-able tubular length 50.

As shown by FIG. 1, circumference closing system 30 provides garment 20 with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system 30 provides garment 20 with an easily adjustable waist portion 22 without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment 20. Although circumference closing system 30 is illustrated as being part of pants 20, circumference closing system 30 may be used in waist portions of other pieces of apparel or other garments.

FIG. 3 illustrates another example piece of apparel or garment 120 which comprises circumference closing system 30. Garment 120 is in the form of a pair of shorts. For purposes of this disclosure, the term "shorts" refers to any garment worn about a person's waist having shorter leg portions than the leg portions 28 of pants 20 or no leg portions. "Shorts" also refers to any garment about a person's waist that have leg openings leg openings that are to be located above the person's ankles, above a person's knees or in some implementations, reach only to a person's thighs. The term "shorts" comprises all forms of shorts including hiking shorts, casual shorts, compression shorts, short tights, swim trunks, capris, skorts and culottes.

As shown by FIG. 3 garment 120 comprises waist portion 22, hips portion 24, crotch portion 26, leg portions 128 and circumference closing system 30. Waist portion 22, hips portion 24, crotch portion 26 and circumference closing system 30 are described above. Leg portions 128 are similar to leg portions 28 except that leg portions 128 are shorter. In the example illustrated, leg portions 128 are sized so as to provide leg openings 38 which are to be located between the thighs and knees of a person wearing garment 120. In other implementations, leg portions 128 may be sized so as to provide leg opening 38 closer to crotch 26 or below a person's knees when garment 120 is worn. In other implementations, one or more of the leg portions 28 or one or more leg openings 38 may include circumference closing system 30.

FIG. 4a illustrates another example piece of apparel or garment 220 which comprises circumference closing system

30. Garment 220 is in the form of a skirt. For purposes of this disclosure, the term “skirt” refers to any garment worn about a person’s waist and generally lacking a crotch or inseam. Examples of different types of skirts include, but not limited to, A-line skirts, broomstick skirts, bubble skirts, circular skirts, dirndl skirt, fishtail skirts, flared skirts, gored skirts, fly skirts, sheath, multi-panel skirts, box pleat skirts, knife pleats skirts, fix-box pleated skirts, prairie skirts, kilts, wrap skirts, jean skirts, tiered skirts, pencil skirts, sarongs, pareos, tulle skirts, handkerchief skirts, trumpet skirts, round skirts, wrap skirts, bazi skirt, cowl skirts, godet skirts and the like.

As shown by FIG. 4a, garment 220 comprises waist portion 22, skirt body 124 and circumference closing system 30. Waist portion 22 and circumference closing system 30 are described above. Skirt body 124 hangs from waist portion 22. As described above, skirt body 124 may be provided in different sizes, shapes and configurations and formed from a single panel or multiple panels of material.

FIG. 4b illustrates another example piece of apparel or garment 270 which comprises circumference closing system 30. Garment 220 is in the form of a pullover jacket. For purposes of this disclosure, the term “jacket” refers to any garment worn about a person’s torso including coats, vests, hoodies, pullovers (which are pulled over a person’s head) and the like. Although FIG. 4b illustrates a pullover jacket, the closing system 30 can be applied to any jacket. The jacket includes a neck region 280, shoulder regions 282, a torso region 286 and arms 288, and an open lower end 272. A constrict-able region 274 can be positioned at the lower end 272 and/or at other locations about jacket. The arms 288 include arm openings 290. For example, a longer jacket can extend below a user’s waist, and may include a first constrict-able region 274 at the waist of the user and/or a second constrict-able region 274 at the open lower end 272 below the waist of the user. Garment 270 includes the circumference closing system at the constrict-able region 274 for selectably adjusting the size of the open lower end and/or the constrict-able region. In other implementations, one or more of the arm openings 290 may include circumference closing system 30.

FIG. 5a illustrates another example garment 320. Garment 320 may have the general form of any of garments 20, 120, 220 and 270 described above except that garment 320 comprises circumference closing system 330. Garment 320 comprises waist portion 22 of a closed system having a continuous waist portion 22 without a separation. In one implementation, waist portion 22 of garment 330 comprises a sleeve 36. In other implementations, sleeve 36 may be omitted. Although circumference closing system 330 is illustrated as being largely contained within waist portion 22, in other implementations, larger portions if not the entirety of circumference closing system 330 may be exposed or external to waist portion 22.

Circumference closing system 330 operates in a somewhat similar fashion to circumference closing system 30 except that circumference closing system 330 utilizes a pair of loops and a pair of lines, wherein adjustment of waist portion 22 occurs through the pair of loops. As shown by FIG. 5a, system 330 comprises loops 332 and lines 340A, 340B (collectively referred to as line 340). Loops 332 are located on a front side of garment 320 along waist portion 22 and are connected to portions of waist portion 22 that collectively extend about the remainder of waist portion 22. In the example illustrated, circumference closing system 330 additionally comprises a band 334 extending within sleeve 36 which has opposite ends secured to loops 332 and extends about the remainder of the interior circumference 34 of waist

portion 22. In other implementations, band 334 may be omitted where loops 332 are affixed to other portions of waist portion 22 or other locations of a garment.

Lines 340 facilitate selected drawing of loops 332 towards one another or the release of such loops 332 to allow movement away from one another so as to thereby adjust the circumference, size or extent of constriction of opening 34 of waist portion 22. Each of lines 340 is substantially similar or identical to one another except that lines 340A, 340B mirror one another on opposite sides of a front centerline of garment 320. Each of lines 340 comprises portions 342, 344, 346, 348 and 350.

Portion 342 comprise an end portion of line 340A, 340B secured to waist portion 22. In one implementation, portion 342 is stitched to waist portion 22. In another implementation, portion 342 is welded, bonded or otherwise secured to waist portion 22. Referring to FIG. 5b, in another implementation, the circumference closing system 330 can be applied to a garment 320 having a separation or break 335 between the portions 342. The separation or break 335 can be a fly of a pant, a full separating jacket hem or other separation.

Referring to FIG. 5a, portion 344 of lines 340A, 340B extends from portion 342 and continues through one of loops 332, within sleeve 36, returning towards portion 342. In the example illustrated, portion 344 comprises a constrict-able tubular length 350 and an exposed portion 352. In the example illustrated, portion 344 comprises a constrict-able tubular length 350 and an exposed portion 352. Constrict-able tubular length 350 comprises a length of line 340A, 340B which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length 350. In one implementation, the entire length of line 340A, 340B is a constrict-able tube. For example, in one implementation, the entire length of line 340A, 340B comprises a braided tube of fibrous material. In another implementation, the entire length of lines 340 comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length 350 may comprise a larger or smaller portion of line 340A, 340B, wherein other portions of lines 340A, 340B are not tubular or are not constrict-able. In still other implementations the constrict-able tubular length 350 can be positioned at any location between portion 342 and the loop 332.

Portion 346 comprises that portion of line 340A, 340B that is extending through the hollow interior of constrict-able tubular length 350, entering at entry 354 and exiting at exit 356. Portion 348 comprises that portion of line 340A, 340B that extends from exit 356 and from length 350. Portion 48 provides a segment of line 340A, 340B by which a person may grasp and pull line 340A, 340B to adjust a length of line 340A, 340B that extends from portion 42, about the waist of the person wearing garment 320, and to the constriction of length 350. In the example illustrated, portion 348 is secured to waist portion 22 at end portion 357. In other implementations, portions 348 may comprise a loose, unattached end.

Exposed portion 352 comprises a segment of portion 344 of line 340A, 340B that is exposed inside of waist portion 22 and sleeve 36 such that a person may grasp portion 352 to pull back portion 346 through length 350 (as indicated by arrows 358), shortening the length of portion 348 and increasing the length of portion 344 extending from the constriction of length 350, through loop 332 and back to portion 342. As the length of portion 348 is pulled back

through length 350, length 350 is in an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when length 350 is in a constricted state), better allowing portion 348 to be pulled back through length 350 towards loop 332. Exposed portion 352 facilitates lengthening of portion 344 and widening of the closable circumference 34 of waist portion 22.

In the example illustrated, sleeve portion 36 and waist portion 22 comprise openings 364, 366, 368 and 370. In one implementation, each of opening 364, 366, 368 and 370 may be defined or protected by a grommet or other lining. In FIGS. 5a and 5d, opening 364 comprises an opening from which portion 344 exits sleeve 36 and waist portion 22. Opening 366 comprises an opening into which portion 344 reenters sleeve 36 and waist portion 22, further extending through loop 332 and back towards constrict-able tubular length 350. Openings 364 and 366 facilitate access to exposed portion 352, allowing line 344 to be loosened when the size of waist portion 22 is to be increased.

Referring to FIG. 5a, opening 368 comprises an opening in sleeve 36 and waist portion 22 through which portion 348 exits, facilitating grasping of portion 348 when waist portion 22 is to be constricted, closed or reduced in size. Opening 370 allows end portion 357 of portion 348 to extend back into waist portion 22 for securement to waist portion 22.

Referring to FIGS. 5a and 5b, when a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22, the person may pull portion 348 through length 350 (as indicated by arrow 360), increasing the length of portion 348 and reducing the length of portion 340A and/or 340B extending from the constriction of length 350, through loop 332 and back to portion 342. After the length of portion 344 and the size of the closable circumference of waist portion 22 have been established, the constriction of portion 350 retains the chosen size of the closable circumference of waist portion 22. Outward forces exerted upon waist portion 22 and upon line portions 344 of lines 340A, 340B place an axial load on length 350 which constricts about and against portion 346. The constriction of portion 350 about portion 346 frictionally locks portion 350 to portion 346 about the entire circumference or on all sides of portion 346, inhibiting axial sliding movement of portion 346 within the hollow interior of constrict-able tubular length 350.

In other implementations end portion 357 could be secured to portion 344 on the opposite side of opening 354 from constrict-able tubular length 350. In FIG. 5b, by grasping portion 348 and applying force in direction 371 toward portion 342, a person can take the axial load off of portion 350, reducing the constriction on portion 346 and lengthening portion 344 which increases the total circumference of waist opening 330. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect. In other implementations in which end portion 357 of portion 348 is unattached, opening 370 may facilitate stowing of the unattached end portion to extend back into waist portion 22 for securement to waist portion 22, if other means of loosening are applied.

Referring to FIG. 5c in one implementation, portion 348 can be divided into tail end portions 341 and 343. End portion 341 passes through opening 370 allowing end portion 357 of portion 341 to extend back into waist portion 22 for securement to waist portion 22. In other implementations end portion 357 could be secured to portion 344 on the opposite side of opening 354 from constrict-able tubular length 350. When portion 348 is divided into tail portions

341 and 343, the tail portion 341 can include a graspable end 359 and the portion 343 can include a graspable end 349. Similar to the implementations of FIGS. 5a and 5b, when a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5c, the person may pull end portion 343 through length 350 (as indicated by arrow 360), increasing the length of portion 343 and reducing the length of portion 340 extending from the constriction of length 350, through loop 332 and back to portion 342. When a person wishes to increase the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5c, a person can grasp end portion 341 and apply force in direction 371 toward 342, which takes the axial load off of portion 350, reduces the constriction on portion 346, and lengthens portion 344 which increases the total circumference of waist opening 34. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

Referring to FIG. 5d, another implementation of the present invention is illustrated with portion 352 providing access for a person to increase the size of closable circumference 34 of waist portion 22, and includes portion 348 as an end portion. When a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5d, the person may pull end portion 348 through length 350 (as indicated by arrow 360), increasing the length of portion 348 and reducing the length of portion 340A extending from the constriction of length 350, through loop 332 and back to portion 342. When a person wishes to increase the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 5d, a person can grasp portion 352 apply force in direction of arrow 358, which takes the axial load off of portion 350, reduces the constriction on portion 346, and lengthens portion 344 which increases the total circumference of waist opening 34. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

As shown by FIGS. 5a through 5d, circumference closing system 330 provides garment 320 with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system 330 provides garment 320 with an easily adjustable waist portion 22 without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment 320. Although circumference closing system 330 is illustrated as being part of pants 20, circumference closing system 30 may be used in other pieces of apparel or other garments. In another implementation, the circumference closing system 330 can be configured with only a single loop 332. In another implementation, the circumference closing system 330 can be configured with only a single line 340A or 340B or with a line 340A or 340B is reduced in size. For example, one of the ends 357 can be directly connected to one of the ends of the band 334 thereby eliminating one of the loops 340A or 340B and/or one of the loops 332.

FIGS. 6a through 6d illustrate another example garments 420. Garment 420 may have the general form of any of garments 20, 120, 220 and 270 described above except that garment 420 comprises circumference closing system 430. Garment 420 comprises waist portion 22. In one implementation, waist portion 22 of garment 420 comprises a sleeve 36. In other implementations, sleeve 36 may be omitted. Although circumference closing system 430 is illustrated as

being largely contained within waist portion 22, in other implementations, larger portions if not the entirety of circumference closing system 430 may be exposed or external to waist portion 22.

Circumference closing system 430 operates in a somewhat similar fashion to circumference closing systems 30 and 330 described above except that circumference closing system 430 utilizes interconnected lines to adjust waist portion 22. As shown by FIG. 6a, system 430 comprises lines 440A, 440B (collectively referred to as lines 440).

Lines 440 facilitate adjustment of the circumference, size or extent of constriction of opening 34 of waist portion 22. Each of lines 440 is substantially similar or identical to one another except that lines 440A, 440B mirror one another on opposite sides of a front centerline of garment 420. Line 440A comprises portions 442A, 444A, 446A and 448A.

Portion 442A comprise an end portion of line 440A secured to waist portion 22. In one implementation, portion 442A is stitched to waist portion 22. In another implementation, portion 442A is welded, bonded or otherwise secured to waist portion 22.

Portion 444A of line 440A extends from portion 442 within sleeve 36 and further extends substantially about the entire circumference 34 of waist portion 22. In the example illustrated, portion 444A comprises a constrict-able tubular length 450A and an exposed portion 452A. In the example illustrated, portion 444A comprises a constrict-able tubular length 450A and an exposed portion 452A. Constrict-able tubular length 450A comprises a length of line 440A which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length 450A. In one implementation, the entire length of line 440A is a constrict-able tube. For example, in one implementation, the entire length of line 440A comprises a braided tube of material. In another implementation, the entire length of line 440A comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length 450A may comprise a portion of line 440A, wherein other portions of line 440A are not tubular or are not constrict-able.

Portion 446A comprises that portion of line 440A that is extending through the hollow interior of constrict-able tubular length 450B, entering at entry 454A and exiting at exit 456A. Portion 448 comprises that portion of line 440A that extends from exit 456A and from length 450B. Portion 448A provides a segment of line 440A by which a person may grasp and pull line 440A to adjust a length of line 440A that extends from portion 442A, about the waist of the person wearing garment 420, and through the constriction of length 450B. In the example illustrated, portion 448A is secured to waist portion 22 at end portion 457A so as to not hang or dangle. In other implementations, portions 448A may comprise a loose, unattached end.

Exposed portion 452A comprises a segment of portion 444A of line 440A that is exposed outside of waist portion 22 and sleeve 36 such that a person may grasp portion 452A to pull back portion 448A through length 450B (as indicated by arrow 458A), shortening the length of portion 448A and increasing the length of portion 444A extending from the constriction of length 450B. As the length of portion 448A is pulled back through length 450B, length 450A releases the axial load on length 450B, returning it to an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when length 450B is in a constricted state), better allowing portion 448B to be pulled

back through length 450B. Exposed portion 452A facilitates lengthening of portion 444A and widening of the closable circumference 34 of waist portion 22. The constricting and release of the closing system 430 can be formed on one or both sides of garment 420 to achieve the desired tightening and/or loosening of the waist portion 22.

Line 440B is similar to line 440A. Line 440B comprises portions 442B, 444B, 446B and 448B. Portion 442B comprise an end portion of line 440B secured to waist portion 22. In one implementation, portion 442B is stitched to waist portion 22. In another implementation, portion 442B is welded, bonded or otherwise secured to waist portion 22.

Portion 444B of line 440B extends from portion 442B within sleeve 36 and further extends substantially about the entire circumference 34 of waist portion 22. In the example illustrated, portion 444B comprises the constrict-able tubular length 450B (noted above) and an exposed portion 452B. Constrict-able tubular length 450B comprises a length of line 440B which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length 450B. In one implementation, the entire length of line 440B is a constrict-able tube. For example, in one implementation, the entire length of line 440B comprises a braided tube of fibrous material such as nylon or polyester. In another implementation, the entire length of line 440B comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length 450B may comprise a shorter or longer portion of line 440B, wherein other portions of line 440B are not tubular or are not constrict-able.

Portion 446B comprises that portion of line 440B that is extending through the hollow interior of constrict-able tubular length 450A, entering at entry 454B and exiting at exit 456B. Portion 448B comprises that portion of line 440B that extends from exit 456B and from length 450A. Portion 448B provides a segment of line 440B by which a person may grasp and pull line 440B to adjust a length of line 440B that extends from portion 442B, about the waist of the person wearing garment 420, and to the constriction of length 450A. In the example illustrated, portion 448B is secured to waist portion 22 at end portion 457B so as to not hang or dangle. In other implementations, portions 448B may comprise a loose, unattached end.

Exposed portion 452B comprises a segment of portion 444B of line 440B that is exposed inside of waist portion 22 and sleeve 36 such that a person may grasp portion 452B to pull back portion 448B through length 450A (as indicated by arrow 458B), shortening the length of portion 448B and increasing the length of portion 444B extending from the constriction of length 450A. As the length of portion 448B is pulled back through length 450A, length 450B releases the axial load on length 450A, returning it to an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when length 450A is in a constricted state), better allowing portion 448B to be pulled back through length 450A. Exposed portion 452B facilitates lengthening of portion 444B and widening of the closable circumference 34 of waist portion 22.

In the example illustrated, sleeve portion 36 and waist portion 22 comprise openings 464A, 464B, 466A, 466B, 468A, 468B and 470A, 470B (collectively referred to as openings 464, 466, 468 and 470, respectively). In one implementation, each of openings 464, 466, 468 and 470 may be defined or protected by a grommet or other lining. Each of openings 464 and 466 comprises an opening from

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which portion 444A, 444B exits sleeve 36 and waist portion 22. Each of openings 466 comprises an opening into which portion 444A, 444B reenters sleeve 36 and waist portion 22. Openings 446 facilitate the provision of exposed portions 452A, 452B, allowing lines 444A, 444B, respectively, to be loosened when the size of waist portion 22 is to be increased. Each of openings 468 comprises an opening in sleeve 36 and waist portion 22 through which portion 448A, 448B exits, facilitating grasping of portion 448A, 448B when waist portion 22 is to be constricted, closed or reduced in size. Each of openings 470 allows the option for end portions 457A, 457B of portions 448A, 448B, respectively, to extend back into waist portion 22 for securement to waist portion 22. In other implementations in which end portions 457 of portions 448 is unattached, opening 470 may facilitate stowing of the unattached end portion.

When a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22, the person may pull on both portions 448. For example, the person may pull portion 448A through length 450B (as indicated by arrow 360A), increasing the length of portion 448A and reducing the length of portion 444A extending about waist portion 22 and back to portion 442A. The person may simultaneously pull portion 448B through length 450A (as indicated by arrow 360B), increasing the length of portion 448B and reducing the length of portion 444B extending about waist portion 22 and back to portion 442A. After the length of portions 444A, 444B and the size of the closable circumference of waist portion 22 have been established, the constriction of portions 450A and 450B retains the chosen size of the closable circumference of waist portion 22. Outward forces exerted upon waist portion 22 and upon line portions 444A, 444B of lines 440A, 440B place an axial load on lengths 450A, 450B which constricts about and against portions 446A, 446B, respectively. The constriction of portions 450 about portions 446 frictionally locks portions 450 to portions 446 about the entire circumference or in all sides of portions 446, inhibiting axial sliding movement of portions 446 within the hollow interior of constrict-able tubular lengths 450.

Referring to FIG. 6b, in another implementation, the circumference closing system 420 can be applied to a garment 420 having a separation or break 435 between the portions 442A and B. The separation or break 435 can be a fly of a pant, a full separating jacket hem or other separation.

Referring to FIGS. 6a and 6b, when a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22, the person may pull portions 448A and B through lengths 450A and B (as indicated by arrows 460A and B), increasing the length of portions 448A and B. After the desired lengths of portions 444A and B and the size of the closable circumference of waist portion 22, have been established, the constriction of portions 450A and B retain the selected size of the closable circumference of waist portion 22. Outward forces exerted upon waist portion 22 and upon line portions 444A and/or B place an axial load on lengths 450A and B which constricts about and against portion 446B and A, respectively. The constriction of portions 450A and B about portions 446B and A, respectively, frictionally locks portion 450A and B to portions 446B and A, respectively, about the entire circumference or on all sides of portions 446A and B, inhibiting axial sliding movement of portion 446B and A within the hollow interior of constrict-able tubular length 450A and B, respectively.

In FIG. 6b, by grasping portion 448A and B, and applying force in direction 471 toward 442A and B, a person can take the axial load off of portions 450B and A, respectively,

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reducing the constriction on portions 446A and B and lengthening portions 444A and B, which increases the total circumference of waist opening 430. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

Referring to FIG. 6c in one implementation, portion 441A passes through opening 470A allowing end portion 457A of portion 444A for securement to waist portion 22. Portion 441A can include a graspable end 459. Similarly, portion 448B can include a graspable end 449. When a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 6c, the person may pull portion 448B through length 450A (as indicated by arrow 460B), increasing the length of portion 448B extending from the constriction of length 450A and reducing the size of the circumference 34. When a person wishes to increase the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 6c, a person can grasp end portion 441A apply force in direction 471 toward portion 442A, which takes the axial load off of portion 450A, reduces the constriction on portion 446B, and lengthens portion 444B which increases the total circumference of waist opening 34. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

Referring to FIG. 6d, another implementation of the present invention is illustrated with portion 452B providing access for a person to increase the size of closable circumference 34 of waist portion 22, and includes portion 448B as an end portion. When a person wishes to reduce or constrict the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 6d, the person may pull end portion 448B through length 450A (as indicated by arrow 460B), increasing the length of portion 448B extending from the constriction of length 450A thereby reducing the length of 444B. When a person wishes to increase the size of closable circumference 34 of waist portion 22 of the implementation of FIG. 6d, a person can grasp portion 452B apply force in direction of arrow 458B, which takes the axial load off of portion 450A, reduces the constriction on portion 446B, and lengthens portion 444B which increases the total circumference of waist opening 34. One would do this on both sides to achieve the maximum circumference, but this could be performed on one side alone for partial effect.

As shown by FIGS. 6a through 6d, circumference closing system 430 provides garment 420 with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system 430 provides garment 420 with an easily adjustable waist portion 22 without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment 420. Although circumference closing system 430 is illustrated as being part of pants 20, circumference closing system 430 may be used in other pieces of apparel or other garments.

FIGS. 7-12 illustrate another example garment 520 comprising a circumference closing system 530. The example is similar to the implementation of FIG. 6d. In the example illustrated, garment 520 comprises a pair of pants or shorts having split circumference 534 (separated at a zipper region or split 535), wherein opposite sides of the split 535 are secured to one another by a clasp or hook 537. In other implementations, split 535 may be omitted. In other implementations, garment 520 may comprise an alternative piece of apparel or garment, such as a skirt or a jacket.

FIGS. 9-12 illustrate circumference closing system 530 in more detail, separated and removed from the remainder of garment 520. As shown by FIG. 9, circumference closing system 530 comprises band 600, guides 602, and lines 540A, 540B (collectively referred to as lines 540). Band 600, guide 602 and lines 540 are substantially contained within the interior of sleeve 536 along waist portion 522.

Band 600 comprises an elongate strap or webbing that extends within sleeve 536 about the circumference 534 of waist portion 522. Band 600 as opposite ends affixed to clasp 537, joining opposite sides of waist portion 522 to form an endless circumference. In the example illustrated, band 600 comprises a strap of elastic material such as polyester or other elastomeric material. The elastic nature of band 600 provide some degree of accommodation for different waist sizes. Moreover, the elastic nature of band 600 distributes loads of lines 540A and B along band 600. In other implementations, band 600 may be inelastic. In yet other implementations, band 600 may be omitted, where lines 540 are contained within sleeve 36 or extend on the exterior waist portion 522.

As shown by FIG. 10, guides 602 comprise channels provided at spaced locations along the length of band 600. Guides 602 provide control passages through which lines 540 pass. Guides 602 maintain lines 540 in close proximity and opposite to band 600 such that changes in the “effective” length of lines 540 results in adjustment of the circumference of band 600 and the circumference 534 of waist portion 522. In other implementations, guide 602 may be omitted or of other configurations.

Line 540A comprises portions 542A, 544A, 546A, 548A and 550A. Portion 542A comprise an end portion of line 540A secured to band 600 in one implementation, portion 542A is stitched to band 600. In another implementation, portion 542A is welded, bonded or otherwise secured to band 600.

Portion 544A of line 540A extends from portion 542A along band 600, guided by guides 602, within sleeve 536 and further extends substantially about the entire circumference 534 of waist portion 522. In the example illustrated, portion 544A comprises a constrict-able tubular length 550A (shown in FIGS. 8 and 11) and an exposed portion 552A (FIG. 8). In the example illustrated, portion 544A comprises a constrict-able tubular length 550A. Constrict-able tubular length 550A comprises a length of line 540A which is in the form of a hollow tube and which constricts or narrows in diameter in response to axial loads placed upon the tube forming constrict-able length 550A. In one implementation, the entire length of line 540A is a constrict-able tube. For example, in one implementation, the entire length of line 540A comprises a braided tube of material. In another implementation, the entire length of line 540A comprises a tube of a resiliently flexible material, such as a resiliently flexible or stretchable elastic material or rubber-like material. In other implementations, constrict-able tubular length 550A may comprise a portion of line 540A, wherein other portions of line 540A are not tubular or are not constrict-able.

Referring to FIGS. 11 and 12, portion 546A comprises that portion of line 540A that is extending through the hollow interior of constrict-able tubular length 550B, entering at entry 554 and exiting at exit 556A where line 540A passes through band 600, forming portion 548A. Portion 548A comprises that portion of line 540A that extends from exit 556A and from length 550B. Portion 548A provides a segment of line 540A by which a person may grasp and pull line 540A to adjust a length of line 540A that extends from

portion 542A, about the waist of the person wearing garment 520, and to the constriction of length 550B. In the example illustrated, portion 548A comprises a loose, unattached end. In other implementations, portion 548A is secured to waist portion 522 so as to not hang or dangle, or to reattach portion 548A to portion 544 adjacent to entry 554 opposing length 550 in order to loosen.

Line 540B is similar to line 540A except that line 540B has portion 542B that has an exposed portion 552B (shown in FIG. 8). Those remaining components of line 540B which correspond to components of line 540A are numbered similarly but followed with a “B” designation rather than an “A” designation.

Exposed portion 552B comprises a segment of portion 544B of line 540A that is exposed inside of waist portion 522 and sleeve 536 such that a person may grasp portion 552B to pull back portion 548B through length 550A (as indicated by arrow 558B), shortening the length of portion 548B and increasing the length of portion 544B extending from the constriction of length 550A. As the length of portion 548B is pulled back through length 550A, tension on length 550B is reduced such that both of lengths 550A and 550B are in an open or un-constricted state (having a larger inner diameter or cross-sectional area as compared to when in a constricted state), better allowing portion 548A to be pulled back through length 550B when a load is applied at 537A. Exposed portion 552B facilitates the widening of the closable circumference 534 of waist portion 522.

In the example illustrated by FIGS. 7 and 8, sleeve portion 536 and waist portion 522 comprise openings 564A, 564B, 566A, 566B and 568A, 568B (collectively referred to as openings 564, 566 and 568, respectively). In one implementation, each of openings 564, 566 and 568 may be defined or protected by a grommet or other lining. Each of openings 564 comprises an opening from which portion 544A, 544B exits sleeve 536 and waist portion 522. Each of openings 566 comprises an opening into which portion 544A, 544B reenters sleeve 536 and waist portion 522. Openings 564 facilitate the provision of exposed portions 552A, 552B, allowing lines 544A, 544B, respectively, to be loosened when the size of waist portion 522 is to be increased. Each of openings 568 comprises an opening in sleeve 536 and waist portion 522 through which portion 548A, 548B exits, facilitating grasping of portion 548A, 548B when waist portion 522 is to be constricted, closed or reduced in size.

When a person wishes to reduce or constrict the size of closable circumference 534 of waist portion 522, the person may pull both of portions 548. For example, the person may pull portion 548A through length 550B (as indicated by arrow 560A), increasing the length of portion 548A and reducing the length of portion 544A extending about waist portion 522 and back to portion 542A. The person may simultaneously pull portion 548B through length 550A (as indicated by arrow 560B), increasing the length of portion 548B and reducing the length of portion 544B extending about waist portion 522 and back to portion 542A. After the length of portions 544A, 544B and the size of the closable circumference of waist portion 522 have been established, the constriction of portions 550A and 550B retains the chosen size of the closable circumference of waist portion 522. Outward forces exerted upon waist portion 522 and upon line portions 544A, 544B of lines 540A, 540B place an axial load on lengths 550A, 550B which constricts about and against portions 546A, 546B, respectively. The constriction of portions 550 about portions 546 frictionally locks portions 550 to portions 546 about the entire circumference or

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on all sides of portions **546**, inhibiting axial sliding movement of portions **546** within the hollow interior of constrictable tubular lengths **550**.

As shown by FIGS. **7** and **8**, circumference closing system **430** provides garment **520** with a user selectable waist portion circumference or size that is reliably retained. Circumference closing system **530** provides garment **520** with an easily adjustable waist portion **522** without requiring the use of a belt, clasps, buttons, buckles, fasteners and the like which might otherwise add undesirable amounts of weight, bulk and/or complexity to the closing system and/or limit movement or flexibility of garment **520**.

Although each of the circumference closing systems **30**, **330**, **430** and **530** have been described as being incorporated into waist portions of various garments such as pants, shorts and skirts, each of the above described circumference closing systems **30**, **330**, **430** and **530** may additionally be incorporated into various other articles that have an adjustable mouth or circumference. For example, each of the above described circumference closing systems **30**, **330**, **430** and **530** may additionally be incorporated into any closed-loop waistband or other article having a circumference to be closed down, such as a waist opening of a jacket, a leg opening of pants, a sleeve opening, a cuff, a hem or the like. In addition to garments such as running shorts, sweatpants, swim shorts, skirts, jackets and the like, such closing systems may be incorporated into bags such as, backpacks, equipment bags, stuff sacks, travel bags, sling bags and other bags having a mouth or opening which can be secured in a closed state, but is openable for the insertion of objects.

FIG. **13** illustrates an example bag **620** incorporating circumference closing system **530** described above. Bag **620** comprises a floor **622** and a mouth **624** having a circumference **534** which may be selectively opened and closed or constricted through the actuation of circumference closing system **530** described above. In such an implementation, circumference closing system **530** retains mouth **624** in a securely closed state. At the same time, pulling of exposed portion **552B** allows mouth **624** to be opened for the insertion or removal of articles.

FIGS. **14-17** illustrate one example method for forming portions of circumference closing system **530**. As illustrated by FIG. **14**, two separate lines **540A** and **540B** are provided. As illustrated by FIG. **15**, a portion of line **540B** is passed through a constrictable tubular portion **550A** of line **540A**. Likewise, a portion of line **540A** is passed through a constrictable tubular portion **550B** of line **540B**. As shown by FIG. **15**, in one implementation, the passing through of one line through to the interior portion of another line may be carried out using a tool **547**.

As illustrated by FIGS. **16** and **17**, end portions **542A** and **542B** are subsequently secured to the garment or other article through stitching or the like, directly or indirectly such as being attached to a band, such as band **600**. As described above, the remaining portions of lines **540A** and **540B** may be contained within the sleeve of the garment or article, wherein at least one portion **552A** and/or **552B** is exposed to allow portions **552A** and/or **552B** to be pulled as indicated by arrows **658A** and **B**, facilitating opening or enlarging the circumference. Portions **548** may be grasped and pulled in the direction indicated by arrows **660** to constrict or close the circumference of the garment or article.

Referring to FIG. **18** through **20**, example implementations of the present invention applied to an article of footwear **800** are illustrated. The article of footwear **800** can be a shoe, a sandal, a boot or any type of footwear. The article **800** of footwear includes an upper **802** and a tongue

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804 defining an opening **806** for receiving the foot of a user, and a plurality of fixed eyelets **808** for routing of a lacing assembly **810**. The eyelets **808** are fixedly coupled to the upper **802** in a spaced apart manner about the perimeter of the opening **806**. In other implementations, the article of footwear can be formed without a tongue. In other implementations, the article of footwear can be formed with other structure or structures for securing the lacing such as, for example, one or more openings, one or more movable eyelets, one or more flexible eyelets, one or more anchors, one or more hooks, one or more latches and combinations thereof.

The lacing assembly **810** includes a first lace **812**, a cord **814** and a floating eyelet **816**. The first lace **812** is an elongate flexible cord or line such as a lacing applied to existing articles of footwear. The first lace **812** has a first lace end **820** coupled to the upper **802** and a second lace end **822**. The cord **814** can be a generally circular braided elongate member including first and second cord ends **824** and **826** and a constricting tube **828** positioned between the first and second cord ends **824** and **826**. The constricting tube **828** can be a braided cord having hollow interior between first and second constricted tube openings **844** and **846**. In other implementations, the cord and the constricted tube can have other constructions. The first lace end **820** is fixedly coupled to the upper **802**, such as, through a stitching. In other implementations, the first lace end **820** can be fixedly coupled to the upper **802** in other manners. The second lace end **822** is then routed through a plurality of the fixed eyelets **808**, the floating eyelet **816**, another one of the eyelets **808**, the constricting tube **828** and an upper fixed eyelet **830** where the second lace end **822** is then available for grasping and pulling by the user. In other implementations, other numbers of eyelets or other fasteners can be used, the order or positioning of the eyelets (fixed and/or movable) and the constricting tube can be varied. The first cord end **824** is fixedly coupled to the upper **802** such as through a stitching. In other implementations, other means for fixing the first cord end to the upper can be used. The second cord end **826** is routed through the floating eyelet **816** and is anchored to the upper **802** and is looped through the upper fixed eyelet **830**. In other implementations, the cord **814** can take alternate paths through one or more eyelets or other fasteners and can be arranged between, before or after the constricting tube.

The portion of the first lace **812** extending through the constricting tube **828** is referred to as a lockdown portion **832**. The portion of the first lace **812** between the first lace end **820** and the constricting tube **828** is referred to as a first lace portion **840**, and the portion of the first lace **812** extending between the constricting tube **828** and the second lace end **822** is the second lace portion **842**. The second lace portion **842** is an adjustable end portion of the first lace **812** in that its length is adjustable based upon which direction the first lace **812** is pulled through the constricting tube **828** by the user.

The lacing assembly **810** provides a durable, reliable lacing for articles of footwear without requiring the use of one or more knots, or one or more fasteners or other hardware. The lacing assembly **810** uses the application of an axial load onto the braided cord **814** when the first lace **812** is tightened. The axial load is accomplished by fixing the first cord end **824** to a first location on the upper **802**, routing the cord **814** through the floating eyelet **816**, and then fixing the second cord **826** at a second location of the upper **802** typically closer to an upper portion of the shoe opening **806**. The first lace **812** can follow a traditional

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lacing path from the first lace end **820** fixed to the upper **802** to the floating eyelet **816**, where the lace **812** extends to another of the eyelets **808** before entering the constricting tube **828** through the first opening **844**. The first lace **812** extends through the constricting tube **828** and exits the tube **828** at the second opening **846** where it extends through the upper fixed eyelet **830**. The orientation of the upper fixed eyelet **830** near or in line with the second opening **846** of the constricting tube **828** protects the second opening **846** of the constricting tube **828** from damage, wear or pulling out from pulling the second portion **842** of the first lace **812**.

The lacing assembly **810** is then tightened by pulling on the second lace portion **842** of the first lace **812** (such as in the direction of arrow **848**). The pulling increases the load on the fixed upper eyelet **830** and the floating eyelet **816** creating an axial load on the braided cord **814** at the constricting tube **828** which causes the constricting tube to constrict, draw down or lock down onto the lockdown portion **832** of the lace **812** extending through the constricting tube **828**. The constricting, clamping or locking down of the constricting tube **828** of the cord **814** onto the lockdown portion **832** provides a secure fixed clamp or lock of the lace **812** with the cord **814** securing the lace **812** in place and keeping the article of footwear **800** securely on the user's foot. In one implementation, the article of footwear **800** may include a pouch **850** or pocket for storing the free second end **822** of the lace **812** after the lace system **810** has been tightened by pulling on the second portion **842** of the lace **812**.

The lacing system **810** can be loosened by pulling on the first lace portion **840** (such as in the direction of arrow **852**). Pulling of the first portion **840** of the first lace **812** reduces the axial load on the constricting tube **828** thereby releasing or lessening of the constriction, clamping or locking-down of the portion **832** of the first lace **812** within the constricting tube **828** enabling the lace **812** to move relative to the constricting tube **828**.

Referring to FIG. 19, lacing system **910** is illustrated. The lacing system **910** is another implementation of the present invention and is substantially similar to the lacing system **810** except that lacing system **910** includes first and second cords **814A** and **814B** positioned on opposite sides of the opening **806** on the upper **802**, first and second floating eyelets **816A** and **816B**, and first and second upper eyelets **830A** and **830B**. Additionally, rather than being fixed to the upper as in system **810**, the first lace end **820** is not fixed or anchored to the upper, but it is routed through the eyelets **808**, the second constricting tube **828B** and the second upper eyelet **830B** to form a first adjustable end **820**. Accordingly, the lacing system **910** provides first and second adjustable ends **820** and **822**, which can be selectively positioned within the pocket **850**.

The lacing system **910** is a double locking system due to the first and second constricting tubes **828A** and **828B**. The lacing system **910** can be tightened by pulling the first and second lace ends **820** and **822**, such as in the directions of **848A** and **848B**. Pulling on lace ends **820** and **822** increases the axial load or tension on the constricting tubes **828A** and **828B** causing the constricting tubes **828A** and **828B** to clamp down, constrict or lockdown on to the portions **832A** and **832B**, respectively, of the lace **812**. In order to loosen the lacing system **910**, a user can pull the portion of the lace **812** between the first and second constricting tubes **828A** and **828B**. The lace system **910** can further include a loosening ring **860** to facilitate the grasping and pulling of the lace **812** from the through the first openings **844A** and **844B** of the constricting tubes **828A** and **828B**. In other

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implementations, the loosening ring can take other shapes or configuration such as including a tab for facilitating the grasping of the loosening ring. The lacing system **910** provides a tight secure closure or tightening of the article of footwear **800** about the user's foot.

FIG. 20 illustrates another implementation of the present invention. In particular, lacing system **1010** is illustrated. Lacing system **1010** is similar to lacing system **910** except that the lace **812** can be longer and the article of footwear **800** can be taller extending over more of the user's foot such as a boot or high top casual shoe. The article of footwear **800** can include a plurality of hooks **862** for facilitating the lacing of the lace **812** about the upper portion of the article of footwear **800**. In other implementations, other fastening mechanisms can be used to secure the lacing through the upper portion of the footwear article, such as, for example, eyelets, openings, loops, other fasteners, and combinations thereof.

Although the present disclosure has been described with reference to example implementations, workers skilled in the art will recognize that changes may be made in form and detail without departing from the spirit and scope of the claimed subject matter. For example, although different example implementations may have been described as including one or more features providing one or more benefits, it is contemplated that the described features may be interchanged with one another or alternatively be combined with one another in the described example implementations or in other alternative implementations. Because the technology of the present disclosure is relatively complex, not all changes in the technology are foreseeable. The present disclosure described with reference to the example implementations and set forth in the following claims is manifestly intended to be as broad as possible. For example, unless specifically otherwise noted, the claims reciting a single particular element also encompass a plurality of such particular elements. The terms "first", "second", "third" and so on in the claims merely distinguish different elements and, unless otherwise stated, are not to be specifically associated with a particular order or particular numbering of elements in the disclosure.

What is claimed is:

1. An article comprising:

a flexible perimeter defining a closable circumference; and

a circumference closing system comprising:

a first line, the first line comprising:

a first portion of the first line secured to the closable circumference at a first location;

a second portion of the first line having a first constrict-able tubular length with a first entry location and a first exit location;

a third portion of the first line extending from the second portion along and in a first direction about the closable circumference; and

a second line, the second line comprising:

a first portion of the second line secured to the circumference at a second location;

a second portion of the second line having a second constrict-able tubular length with a second entry location and a second exit location;

a third portion of the second line extending from the second portion along and in a second direction, opposite the first direction, about the closable circumference,

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wherein the first line further comprises:

a fourth portion of the first line extending from the third portion into the second entry location and through the second constrict-able tubular length; and

a fifth portion of the first line exiting the second constrict-able tubular length at the second exit location, and

wherein the second line further comprises:

a fourth portion of the second line extending from the third portion of the second line into the first entry location and through the first constrict-able tubular length; and

a fifth portion of the second line exiting the first constrict-able tubular length at the first exit location of the first line.

2. The article of claim 1, wherein the fifth portion of the first line and the fifth portion of the second line are secured to the article.

3. The article of claim 1, wherein an entire length of the first line and an entire length of the second line are each a constrict-able tube.

4. The article of claim 1, wherein each constrict-able tube comprises a braided tube.

5. The article of claim 1, wherein the first constrict-able tubular length and the second constrict-able tubular length each comprise a braided constrict-able tubular length.

6. The article of claim 1, wherein the article comprises a sleeve comprising first and second gaps and first and second openings, wherein the first constrict-able tubular length extends within the sleeve between the first location and the first gap, wherein the second constrict-able tubular length extends within the sleeve between the second location and the second gap, wherein the third portion of the first line extends across the second gap and wherein the third portion of the second line extends across the first gap.

7. A bag comprising:

a floor;

a mouth providing access to an interior of the bag; and

a mouth closing system comprising:

a first line, the first line comprising:

a first portion of the first line secured to the mouth at a first location;

a second portion of the first line having a first constrict-able tubular length with a first entry location and a first exit location;

a third portion of the first line extending from the second portion along and in a first direction about the mouth; and

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a second line, the second line comprising:

a first portion of the second line secured to the mouth at a second location;

a second portion of the second line having a second constrict-able tubular length with a second entry location and a second exit location;

a third portion of the second line extending from the second portion along and in a second direction, opposite the first direction, about the mouth,

wherein the first line further comprises:

a fourth portion of the first line extending from the third portion into the second entry location and through the second constrict-able tubular length; and

a fifth portion of the first line exiting the second constrict-able tubular length at the second exit location, and

wherein the second line further comprises:

a fourth portion of the second line extending from the third portion of the second line into the first entry location and through the first constrict-able tubular length; and

a fifth portion of the second line exiting the first constrict-able tubular length at the first exit location of the first line.

8. The bag of claim 7, wherein the fifth portion of the first line and the fifth portion of the second line are secured to the bag.

9. The bag of claim 7, wherein an entire length of the first line and an entire length of the second line are each a constrict-able tube.

10. The bag of claim 7, wherein each constrict-able tube comprises a braided tube.

11. The bag of claim 7, wherein the first constrict-able tubular length and the second constrict-able tubular length each comprise a braided constrict-able tubular length.

12. The bag of claim 7, wherein the bag comprises a sleeve comprising first and second gaps and first and second openings, wherein the first constrict-able tubular length extends within the sleeve between the first location and the first gap, wherein the second constrict-able tubular length extends within the sleeve between the second location and the second gap, wherein the third portion of the first line extends across the second gap and wherein the third portion of the second line extends across the first gap.

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